

SyllCraft — User Manual

Version 1.0.0

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1. Introduction

SyllCraft is a free, open-source educational tool that automatically divides English words into syllables. It was designed specifically for Brazilian learners of English as a Foreign Language (EFL) and for teachers in training, acknowledging that the opacity of English orthography — its non-transparent letter-to-sound correspondence — is a systematic source of difficulty for speakers of Brazilian Portuguese (BP), a language with much more regular grapheme-phoneme mappings.

Rather than providing a single "correct" division, SyllCraft presents three independent analyses side by side:

- **Orthographic** — how the word is hyphenated in American English dictionaries
- **Phonetic (V-V)** — how the word divides phonetically, based on vowel onsets (ambisyllabic model)

- **Phonological (SSP-MOP)** — how the word divides according to universal phonological principles (Maximal Onset Principle and Sonority Sequencing Principle)

This multi-criteria display is intentional: it encourages learners to reflect on the differences between writing and sound, and between phonetic and phonological levels of analysis, cultivating the *meta-phonetic-phonological awareness* that is essential for L2 pronunciation development.

2. System Requirements

Component	Minimum requirement
Operating system	Windows 10, macOS 10.14, or any modern Linux distribution
Python	Version 3.7 or higher
Disk space	< 50 MB (including dependencies)
Display	Any screen resolution; GUI dialog is 600 × 370 px
Internet	Required only for initial pip installation of dependencies

3. Installation

3.1 Installing Python

If Python is not already installed on your system, download and install it from the official website:

<https://www.python.org/downloads/>

During installation on Windows, check the box **"Add Python to PATH"**.

Verify your installation by opening a terminal (Command Prompt on Windows, Terminal on macOS/Linux) and typing:

```
python --version
```

You should see output such as **Python 3.11.0** or similar.

3.2 Setting up a virtual environment (recommended)

Using a virtual environment prevents conflicts with other Python projects on your system.

Create a virtual environment:

```
python -m venv syllcraft_env
```

Activate the virtual environment:

- On Windows:

```
syllcraft_env\Scripts\activate
```

- On macOS / Linux:

```
source syllcraft_env/bin/activate
```

You will see the environment name appear in parentheses in your terminal prompt, indicating that the environment is active.

3.3 Installing dependencies

With your virtual environment active (or directly in your system Python), install the required packages:

```
pip install eng-to-ipa pyphen tabulate
```

This installs three packages:

- **eng-to-ipa**: Converts English words to their IPA (International Phonetic Alphabet) transcription.
- **pyphen**: Performs orthographic hyphenation using Hunspell dictionaries (American English).
- **tabulate**: Formats and prints the results as a structured terminal table.

The fourth dependency, **tkinter**, is part of the Python standard library and does not need to be installed separately. On some Linux distributions, you may need to install it via your system package manager:

```
# Debian / Ubuntu
```

```
sudo apt-get install python3-tk
```

```
# Fedora / Red Hat
```

```
sudo dnf install python3-tkinter
```

4. Launching SyllCraft

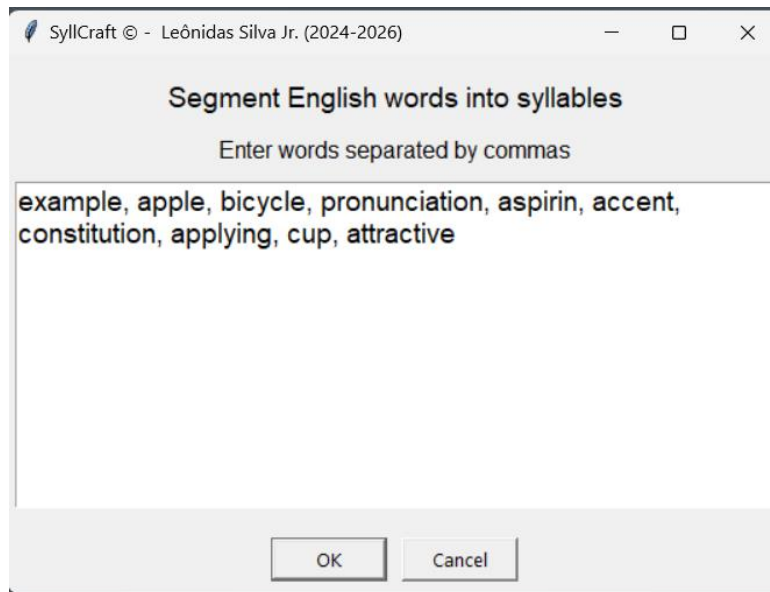
Navigate to the folder where `SyllCraft.py` is saved and run:

```
python SyllCraft.py
```

The graphical input dialog will open immediately.

5. The Input Dialog

When SyllCraft launches, a dialog window titled "**SyllCraft © — Leônidas Silva Jr. (2024–2026)**" will appear.



Instructions:

1. Clear the default text in the text area, or modify it as needed.
2. Type your English words, **separated by commas**. For example:

foreign, rhythm, constitution, bicycle, pronunciation
3. Click **OK** to run the analysis. Click **Cancel** or close the window to exit.

There is no practical limit on the number of words you can enter in a single session. The text area is scrollable for long lists.

6. Understanding the Output Table

After clicking OK, SyllCraft prints a formatted table in the terminal. Each row corresponds to one input word, and each column represents a different segmentation criterion.

```
===== Ambisyllabification (Selkirk, 1982, 1984; Clements, 1990; Barbosa, 2006; Ladefoged & Johnson, 2011) =====
```

Word	Ortho	Ortho_Syll	Phon	Phon_Syll	SSP-MOP	SSP-MOP_Syll
example	['ex', 'ample']	2	['ɪgz', 'əmp', 'əl']	3	['ɪg', 'zəm', 'pəl']	3
apple	['apple']	1	['æp', 'əl']	2	['æ', 'pəl']	2
bicycle	['bi', 'cycle']	2	['baɪs', 'ɪk', 'əl']	3	['baɪ', 'sɪ', 'kəl']	3
pronunciation	['pro', 'nun', 'ci', 'ation']	4	['prən', 'əns', 'i', 'eɪ', 'ən']	5	['prə', 'nən', 'si', 'eɪ', 'fən']	5
aspirin	['as', 'pi', 'in']	3	['æspɪr', 'ɪn']	2	['æsp', 'rɪn']	2
accent	['ac', 'cent']	2	['æks', 'ɛnt']	2	['æk', 'sɛnt']	2
constitution	['con', 'sti', 'tu', 'tion']	4	['kɒnst', 'ət', 'u', 'ən']	4	['kɒns', 'tə', 'tu', 'fən']	4
applying	['ap', 'ply', 'ing']	3	['æpl', 'aɪ', 'ɪŋ']	3	['æp', 'laɪ', 'ɪŋ']	3
cup	['cup']	1	['kʌp']	1	['kʌp']	1
attractive	['at', 'tract', 'ive']	3	['atr', 'ækt', 'ɪv']	3	['æt', 'ræk', 'tɪv']	3

6.1 Orthographic segmentation (Ortho)

This column shows how the word would be hyphenated in American English according to the Hunspell dictionary — the same standard used by major American dictionaries (e.g., Merriam-Webster) and word processing software.

This segmentation is based on **written conventions**, not on phonetic reality. It is the criterion most familiar to learners who have been taught to divide words orthographically. Because English orthography is opaque, this segmentation will sometimes diverge from the phonetic and phonological analyses.

Column headers:

- **Ortho** — list of orthographic syllable strings
- **Ortho_Syll** — total count of orthographic syllables

6.2 Phonetic segmentation (Phon / V-V model)

This column implements the **phonetic syllable** as a V-V interval: the stretch of speech from the onset of one vowel to the onset of the next vowel (Barbosa, 2006). Each syllable is displayed in IPA transcription.

Inter-vocalic consonants are treated as **ambisyllabic** (Ladefoged & Johnson, 2011): they are attached to the onset of the following vowel rather than strictly assigned to one syllable or the other. This reflects the perceptual-articulatory reality of English speech, in which consonants between vowels are not strictly bounded.

This model is grounded in the concept of **perceptual centers (P-centers)** (Morton et al., 1976), which locates the psychological moment of a syllable's occurrence near the onset of its vowel, and in the articulatory oscillator model of speech rhythm (Barbosa, 2006).

Column headers:

- **Phon** — list of phonetic syllables in IPA
- **Phon_Syll** — total count of phonetic syllables

6.3 Phonological segmentation (SSP-MOP)

This column implements the phonological syllabification algorithm of Selkirk (1982, 1984), integrating the **Maximal Onset Principle (MOP)** and the **Sonority Sequencing Principle (SSP)** (Clements, 1990). Results are displayed in IPA.

Maximal Onset Principle (MOP): When a consonant appears between two vowels, it is assigned to the onset of the following syllable — maximizing the onset — provided this does not violate English phonotactic constraints. For example, *data* segments as /dej.tə/ (not */dejt.ə/), because /t/ can legally begin a syllable in English.

Sonority Sequencing Principle (SSP): Within any syllable, sonority must rise from the onset toward the nucleus (typically a vowel) and fall from the nucleus to the coda. The sonority hierarchy runs approximately: vowels > glides > liquids > nasals > fricatives > stops. A syllable onset like /stu/ (as in *string*) is valid because sonority rises from stop to fricative to liquid toward the vowel.

Column headers:

- **SSP-MOP** — list of phonological syllables in IPA
- **SSP-MOP_Syll** — total count of phonological syllables

7. The Output File

In addition to the terminal table, SyllCraft automatically writes the results to a file named **segmented_words.txt** in the same directory from which the script was run. This is a **tab-delimited, UTF-8 encoded text file** with one word per row.

File structure

Column	Content
Word	The input word
Ortho	Orthographic syllables joined by hyphens
Ortho_Syll	Number of orthographic syllables
Phon	Phonetic syllables (V-V) joined by hyphens
Phon_Syll	Number of phonetic syllables
SSP-MOP	Phonological syllables joined by hyphens
SSP-MOP_Syll	Number of phonological syllables

Opening the file

- **Microsoft Excel / LibreOffice Calc:** Open the file and select "Tab" as the delimiter when prompted.
- **R:** Use `read.table("segmented_words.txt", sep="\t", header=TRUE, fileEncoding="UTF-8")`.
- **Any text editor:** The file is plain UTF-8 text and can be read directly.

8. Worked Examples

Below are representative examples illustrating how the three criteria can agree or disagree, and what pedagogical insight each comparison provides.

Example 1: *apple*

Criterion	Result	Syllables
Orthographic	ap-ple	2
Phonetic V-V	æ-pəl	2
SSP-MOP	æ-pəl	2

All three criteria agree on two syllables. The orthographic form shows a doubled consonant (pp), while both phonetic analyses, based on IPA, simplify this to a single /p/ at the onset of the second syllable.

Example 2: *foreign*

Criterion	Result	Syllables
Orthographic	for-eign	2
Phonetic V-V	fɔɹn	1
SSP-MOP	fɔɹn	1

This example is highly instructive for Brazilian learners: the word has three vowel *letters* (o, e, i) but only one vowel *sound* /ɔ/. The orthographic method (based on written conventions) returns two syllables, while both phonetically grounded methods correctly identify the word as a monosyllable.

Example 3: *constitution*

Criterion	Result	Syllables
Orthographic	con-sti-tu-tion	4
Phonetic V-V	kən-stɪ-tu-fən	4
SSP-MOP	kən-stɪ-tu-fən	4

All three agree on four syllables. The IPA output in the phonetic columns allows learners to compare the written form with the actual sounds, explicitly exposing mismatches such as the *ti* → /f/ correspondence.

Example 4: *rhythm*

Criterion	Result	Syllables
Orthographic	rhyth-m	2
Phonetic V-V	ɹɪ-ðm	2
SSP-MOP	ɹɪ-ðm	2

A challenging word for learners because it contains a syllabic consonant (*m*). All three methods agree on two syllables, but the IPA representation reveals that the second syllable has no vowel — a structure very unusual in Brazilian Portuguese.

9. Pedagogical Guidelines

SyllCraft is most effective when used in structured, explicit instruction sessions. The following guidelines are based on the classroom protocol validated at the State University of Paraíba (UEPB):

Preparation

- Pre-select a set of 15–25 words varying in syllabic complexity. Include words that are both phonetically transparent (e.g., *cup*, *bug*) and opaque (e.g., *foreign*, *rhythm*, *thorough*).
- Ask learners to perform a segmentation task manually — by hand, without the tool — before using SyllCraft. This establishes a baseline and primes reflection.

During instruction

- Run the selected words through SyllCraft and project the output table for the class.
- Direct learners to compare columns: Where does the orthographic segmentation differ from the phonetic one? Why?
- Explicitly discuss the concepts of onset, nucleus, coda, the MOP, and the SSP using the IPA output as a concrete visual anchor.
- Ask learners to record differences between their manual segmentation and the tool's output, and to justify any disagreement.

After instruction

- Have learners repeat the manual segmentation task with a new set of words of equivalent complexity.
- Compare pre- and post-instruction performance to measure metalinguistic gains.

Key concepts to reinforce through the tool:

- The distinction between orthographic letters and phonological segments
- The role of the vowel as the nucleus (obligatory syllable core) in English
- Why consonant clusters are assigned to onset positions (MOP)
- How sonority constraints explain which clusters are phonotactically legal
- The notion of ambisyllabicity and why some consonants resist strict assignment

10. Known Limitations

- **IPA transcription accuracy:** SyllCraft uses `eng_to_ipa` for automatic transcription, which reflects a standard American English pronunciation model. Transcription errors may occur for low-frequency or irregular words.

- **Phonotactic implementation:** The current SSP-MOP implementation applies a simplified consonant inventory and sonority hierarchy. Complex consonant clusters (e.g., three-consonant onsets in /stɹ/) may not always be segmented identically to what a trained phonologist would produce.
- **Language coverage:** Version 1.0.0 is exclusively for English. Future versions may incorporate additional languages.
- **Morphological boundaries:** The tool does not currently analyze morphological structure (prefixes, suffixes, roots). Syllabification of derived forms may occasionally conflict with morphological expectations.
- **Proper nouns and abbreviations:** These are not handled reliably by the IPA transcription backend.

11. Troubleshooting

The dialog does not open / tkinter error On some Linux systems, tkinter must be installed separately. Run:

```
sudo apt-get install python3-tk
```

Then re-run `python SyllCraft.py`.

ModuleNotFoundError: No module named 'eng_to_ipa' (or `pyphen`, `tabulate`) Install the missing package:

```
pip install eng-to-ipa pyphen tabulate
```

The output file `segmented_words.txt` is not created Ensure you have write permission in the directory where SyllCraft is running. Try running it from your home directory or desktop.

IPA output contains * or garbled symbols This occurs when `eng_to_ipa` cannot recognize a word (e.g., proper nouns, non-English words). Check the spelling of your input, or substitute a standard English dictionary form.

The table appears misaligned in the terminal Make sure your terminal uses a monospaced font. On Windows, use Windows Terminal or PowerShell with a Nerd Font. On macOS/Linux, any standard terminal emulator should display the table correctly.

12. References

- Barbosa, P. A. (2006). *Incursões em torno do ritmo da fala*. Campinas: Pontes Editores.
- Clements, G. N. (1990). The role of the sonority cycle in core syllabification. In J. Kingston & M. E. Beckman (Eds.), *Papers in Laboratory Phonology* (pp. 283–333). Cambridge: Cambridge University Press.

Ladefoged, P., & Johnson, K. (2011). *A course in phonetics*. Boston: Wadsworth Cengage Learning.

Morton, J., Marcus, S., & Frankish, C. (1976). Perceptual centers (P-centers). *Psychological Review*, 83(5), 405–408. <https://doi.org/10.1037/0033-295X.83.5.405>

Selkirk, E. (1982). The syllable. In H. Houst & V. D. Smith (Eds.), *The structure of phonological representation* (Part 1). Berlin/Boston: De Gruyter. <https://doi.org/10.1515/9783112328088>

Selkirk, E. (1984). *Phonology and syntax: The relation between sound and structure*. Cambridge: MIT Press.

Silva Jr., L. J. (2024–2026). *SyllCraft* (Version 1.0.0) [Computer software]. <https://github.com/leonidasjr/SyllCraft>